

DES MOINES 17 E, IA, USA

WMO: 744480

Lat: 41.5562N

Lon: 93.2855W

Elev: 921

StdP: 14.21

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54902

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	-1.1	-11.9	3.0	-5.3	-7.5	3.8	-0.2	23.9	19.0	20.8	21.3	5.6	N/A	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	90.6	76.0	88.1	75.4	85.5	74.0	79.1	87.2	77.2	84.8	75.4	83.2	6.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	144.9	85.2	74.9	135.6	82.6	73.0	127.0	80.7	43.4	87.2	41.3	84.9	39.6	83.3	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	16.3	14.1	-13.9	94.6	5.5	3.6	-17.9	97.2	-21.1	99.3	-24.1	101.4	-28.1	104.0		
			-14.6	82.1	5.7	1.2	-18.7	83.0	-22.1	83.7	-25.3	84.4	-29.5	85.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	21.8	24.3	38.7	50.8	61.3	70.8	74.1	71.8	65.4	52.2	39.3	26.8
	DBStd	20.82	12.87	12.65	12.02	9.87	8.31	5.42	5.67	5.13	7.88	9.39	10.55	11.48
	HDD50	3258	873	722	390	108	11	0	0	0	2	86	346	720
	HDD65	6377	1338	1140	818	434	170	12	4	6	90	409	772	1184
	CDD50	3228	0	2	39	132	362	625	748	677	464	155	23	1
	CDD65	873	0	0	2	8	56	187	286	218	102	14	0	0
	CDH74	7464	0	0	20	113	452	1531	2630	1734	863	119	2	0
	CDH80	2352	0	0	2	22	105	452	962	523	264	22	0	0
Wind	WSAvg	6.0	7.4	7.3	7.4	8.1	6.5	4.9	4.0	3.7	4.5	5.8	6.3	6.6
Precipitation	PrecAvg	36.0	0.9	1.1	2.1	3.8	5.0	5.6	4.8	4.3	3.6	2.3	1.9	1.1
	PrecMax	51.5	2.0	2.1	5.9	8.0	8.4	12.3	11.7	11.9	6.9	6.3	5.3	2.9
	PrecMin	27.2	0.4	0.2	0.0	1.4	1.0	1.3	0.9	0.5	1.6	0.3	0.0	0.1
	PrecStd	6.6	0.4	0.6	1.5	1.6	1.8	2.8	2.9	2.8	1.5	1.6	1.3	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	63.6	77.1	83.6	88.6	90.8	96.2	93.0	90.2	83.4	71.5	57.9
		MCWB	44.2	51.4	63.2	65.8	73.0	76.5	77.0	73.9	73.4	67.2	57.8	49.5
	2%	DB	47.7	53.9	69.6	77.0	82.6	88.1	91.7	88.6	86.7	77.2	65.1	51.1
		MCWB	42.2	45.1	57.1	61.4	68.9	75.4	77.6	74.9	73.6	63.3	55.0	45.8
	5%	DB	42.7	46.9	63.3	72.3	78.9	85.2	88.7	85.5	82.9	72.7	59.7	47.2
		MCWB	37.9	41.4	53.5	58.4	66.0	73.6	76.9	74.0	70.9	61.6	50.9	42.6
	10%	DB	38.4	41.4	57.1	67.7	75.5	82.3	85.4	82.7	78.9	68.1	55.1	42.0
		MCWB	35.2	37.4	49.5	55.1	64.9	71.9	75.7	72.4	68.2	59.4	47.8	38.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	55.3	64.0	66.9	74.7	80.6	81.7	80.6	76.9	70.3	59.4	53.6
		MCDB	53.0	61.8	75.5	79.2	86.3	88.9	90.7	88.3	87.0	75.5	66.2	55.8
	2%	WB	41.4	47.2	59.8	63.1	71.5	77.8	79.6	77.3	74.9	66.8	55.9	48.0
		MCDB	46.9	53.4	66.6	74.0	81.4	85.8	87.9	83.8	84.2	72.0	63.3	49.8
	5%	WB	38.1	41.9	54.9	60.1	69.3	75.4	77.9	75.2	73.1	63.8	51.8	42.6
		MCDB	41.8	47.5	61.9	69.9	77.1	82.4	85.6	82.9	80.8	68.5	57.9	47.4
	10%	WB	35.3	37.5	49.8	57.0	67.0	73.4	76.2	73.5	71.0	59.8	48.3	38.6
		MCDB	38.3	41.7	57.3	65.0	73.7	80.6	83.7	80.8	76.8	66.9	54.1	42.2
Mean Daily Temperature Range	5% DB	MDBR	16.2	17.3	19.1	21.1	19.1	18.5	19.0	19.1	21.1	19.9	18.3	15.6
		MCDBR	21.4	24.6	27.0	27.9	22.8	21.0	21.8	21.7	23.5	24.9	24.9	21.1
		MCWBR	15.9	16.9	16.3	15.9	11.9	11.7	10.3	11.0	10.9	13.6	15.7	14.8
	5% WB	MCDBR	20.3	24.9	24.0	24.8	20.6	19.2	18.4	19.3	19.4	19.7	23.2	19.0
		MCWBR	15.4	18.0	14.7	15.1	11.6	12.2	10.6	10.6	10.2	13.5	16.6	14.6
Clear-Sky Solar Irradiance	taub	0.290	0.315	0.331	0.383	0.409	0.421	0.438	0.419	0.385	0.343	0.301	0.290	
	taud	2.416	2.346	2.396	2.271	2.266	2.279	2.219	2.287	2.335	2.448	2.498	2.432	
	Ebn at Noon	274	282	291	281	275	271	265	267	269	270	269	264	
	Edh at Noon	25	31	34	41	42	42	44	40	36	28	23	23	
All-Sky Solar Radiation	RadAvg	609	928	1203	1505	1773	1969	1989	1726	1456	975	662	509	
	RadStd	57	62	113	113	138	193	148	110	108	113	56	48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page